



CY 704: Introduction to Theoretical Chemistry

Course Code	CY 704
Course Name	Introduction to Theoretical Chemistry
L-T-P-C	3-0-0-3
Prerequisites	M.Sc. Semester: Odd
Intended for	Ph.D.
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- The Basics of Quantum Mechanics, Model Problems, Exact Solution. [6 Lectures]
- Understanding Energy surfaces, beyond model problems, normal modes, local modes, transition states, symmetry. [6 Lectures]
- An overview of spectroscopy, theoretical chemistry: Structure, bonding, rate of changes, Molecular [6 Lectures]
- Electronic structure: Structure atomic and molecular orbitals, Experimental Probes of Electronic Structure [6 Lectures]
- Statistical Mechanics: Collections of Molecules at or Near Equilibrium, Monte Carlo Evaluation of Properties, Molecular Dynamics Simulations, applications. [10 Lectures]
- Chemical Dynamics: Theoretical Treatment of Chemical Change and Dynamics, Experimental Probes of Reaction Dynamics. [8 Lectures]

Text and Reference Books

1. Jack Simons, **An Introduction to Theoretical Chemistry**, Cambridge University Press, 2003.
2. C. J. Cramer, **Essentials of Computational Chemistry**, Wiley, 2002.
3. R. D. Levine and R. B. Bernstein, **Molecular Reaction Dynamics and Chemical Reactivity**, Oxford University Press, 1997.
4. A. R. Leach, **Molecular Modeling**, 2nd edition, Prentice Hall, 2001.